



**COMSATS University Islamabad, Attock Campus**  
**Program: BS (CE) – VII**

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**Course: Wireless Communication Systems (EEE464)**

# Wireless Sensor Network Design for Smart Agriculture – Greenhouse Monitoring

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## 1. Specific Application

### Real-World Problem:

Greenhouses require precise control of environmental parameters (temperature, humidity, soil moisture, light intensity, CO<sub>2</sub> levels) to optimize plant growth and resource use. Manual monitoring is labor-intensive and prone to human error.

### Justification for WSN:

- **Real-time monitoring** enables automated control of irrigation, ventilation, and lighting.
- **Early detection** of pest infestations, diseases, or equipment failures.
- **Resource efficiency** by reducing water and energy waste.
- **Scalability** to large greenhouse complexes.

## 2. Network Architecture

### Topology: Hybrid (Cluster-based Mesh)

### Structure:

- Sensor nodes form small clusters (5–10 nodes per cluster) that report to a Cluster Head (CH).
- CHs are interconnected via a mesh to increase resilience and allow multi-hop routing among CHs.
- One or more Gateways/Sinks connect the mesh to the cloud or local control server.

### Justification:

- Clustering reduces redundant uplinks and conserves node energy through aggregation.
- Mesh between CHs adds fault tolerance (alternate paths if one CH fails) and helps coverage in obstructed greenhouse spaces.

- Hybrid design balances energy efficiency, reliability, and scalability.

### 3. Sensor Node Design

#### A. Sensor (End) Node — Typical Specifications

- **Sensing Modules**
  - **Temperature & humidity:** DHT22 or SHT31 (higher accuracy).
  - **Soil moisture:** Capacitive moisture probe (less corrosion).
  - **Light intensity:** BH1750 (I<sup>2</sup>C).
  - **CO<sub>2</sub> / air quality:** MH-Z19 / MQ-135 (select appropriate CO<sub>2</sub> sensor for greenhouse levels).
- **Microcontroller**
  - **ESP32** (chosen) or low-power MCU (STM32L0) if Wi-Fi is undesired. ESP32 provides Wi-Fi + BLE for flexible local connectivity; deep-sleep modes for energy savings.
- **Communication**
  - IEEE 802.15.4 (ZigBee) for low-power mesh and multi-hop inside greenhouse.
  - Optionally LoRa for long-range uplink from CHs to gateway if gateway sits far or through metal/glass attenuation.
- **Power**
  - Primary: AA NiMH or Li-ion battery pack.
  - Optional: small solar panel if greenhouse roof allows recharge (or wired power for permanently mounted nodes).
- **Enclosure**
  - IP65-rated housing, adequate ventilation for air sensors, and probe cabling for soil sensors.

#### B. Cluster Head (CH)

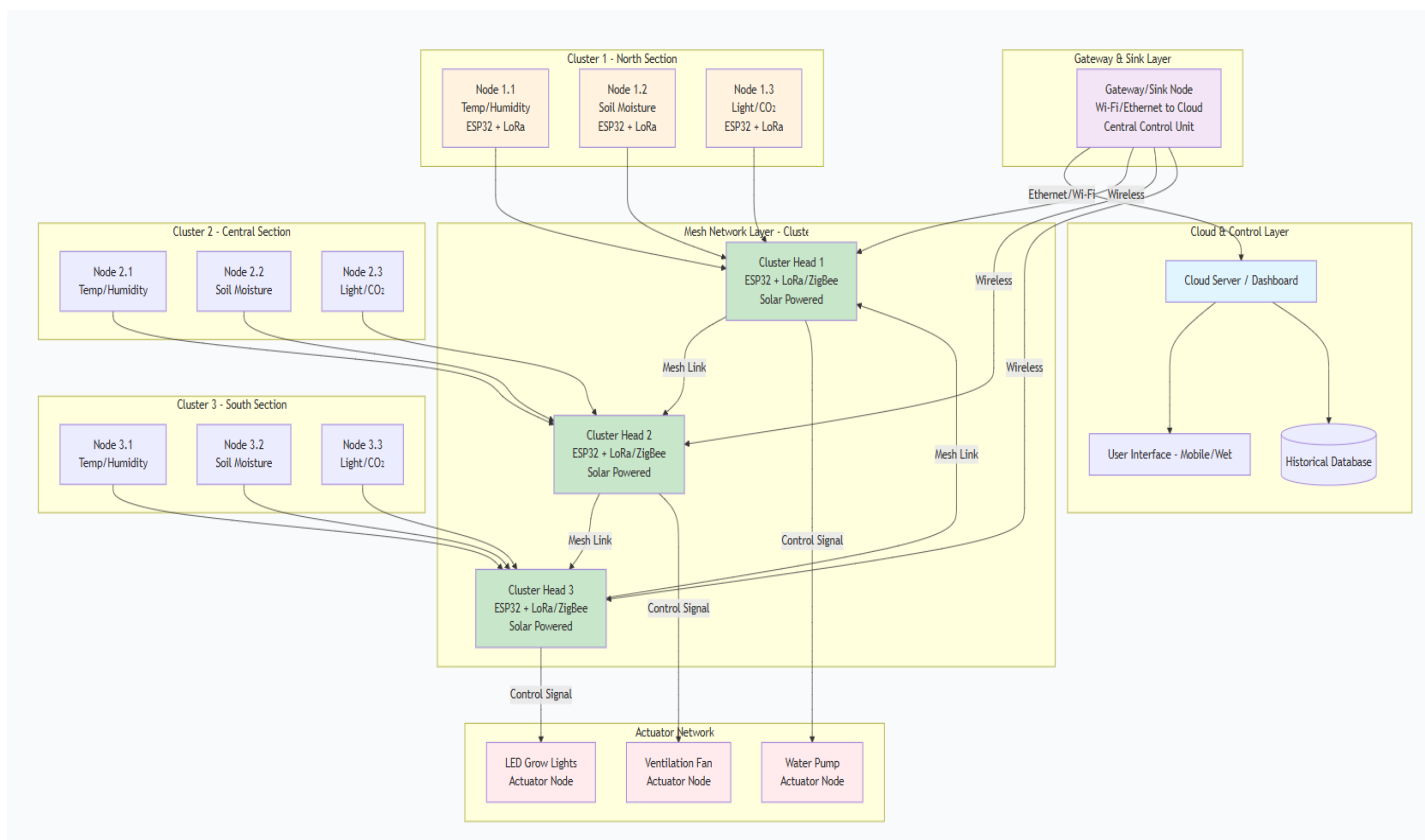
- More powerful MCU or SBC (e.g., ESP32 + external LoRa module or Raspberry Pi Zero for heavy processing).

- Dual-radio: short-range (ZigBee/802.15.4) + long-range (LoRa/Wi-Fi) for gateway uplink.
- Larger battery + solar panel, or mains power where available.
- Actuation outputs: relay drivers or SSRs for pumps, fans, shades.
- Local data buffering (SD card) and OTA firmware update capability.

## 4. Communication and Routing Protocol

- **MAC Protocol: TDMA-based (Time Division Multiple Access)**
  - Reduces collisions and idle listening.
  - Energy-efficient for scheduled data transmissions.
- **Routing Protocol: LEACH (Low Energy Adaptive Clustering Hierarchy)**
  - Cluster heads rotate to balance energy consumption.
  - Data aggregation at cluster heads reduces transmission load to the sink.

### Block Diagram:



- **Sleep Scheduling:** Nodes enter deep sleep between readings (e.g., sample every 10 minutes).
- **Duty Cycling:** Radio is only active during assigned TDMA slots.
- **Energy Harvesting:** Solar panels for cluster heads or gateway nodes.
- **Adaptive Sensing:** Reduce sampling rate when environmental conditions are stable.

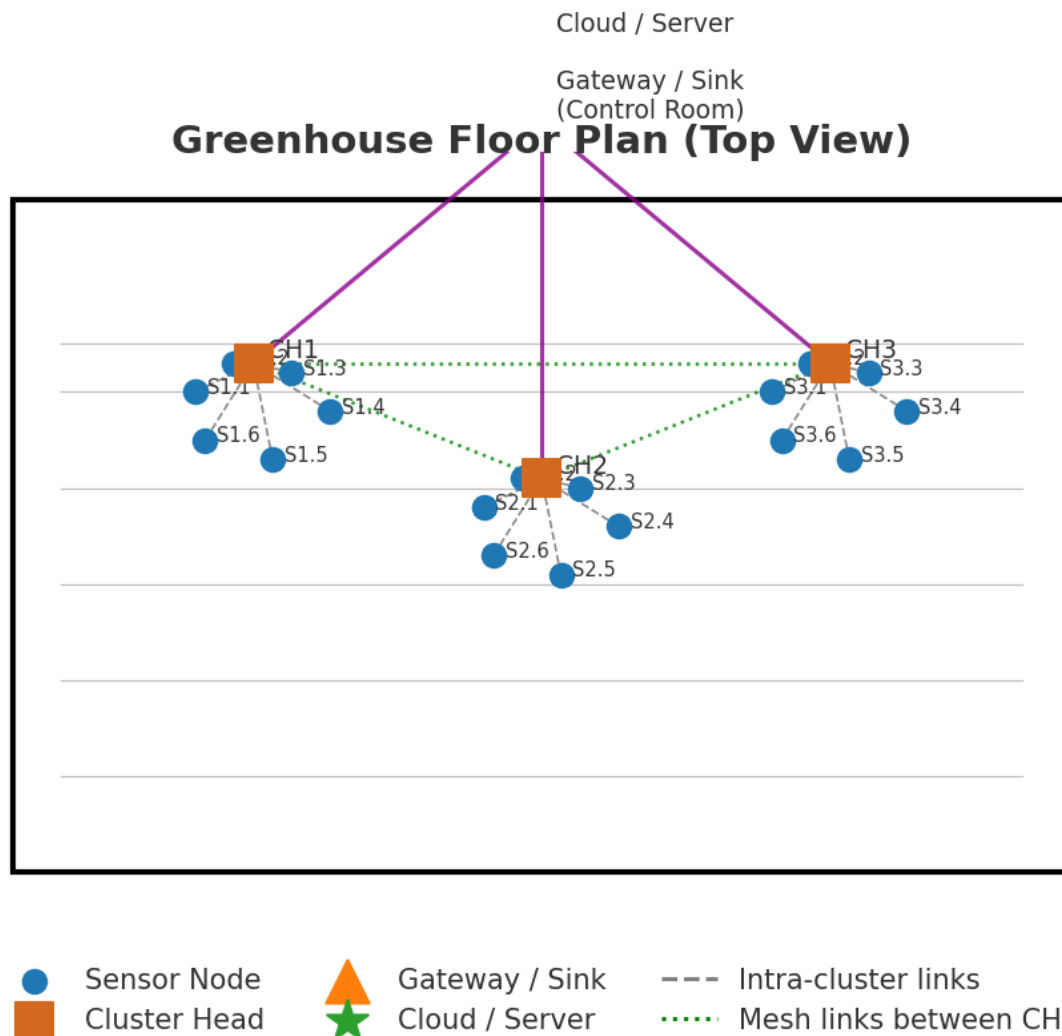
## 6. Data Collection and Sink Node Placement

**Sink / Gateway placement:** centrally in greenhouse near control room or on elevated mounting to reduce attenuation. Use wired power/internet for gateway if possible.

**Data flow:**

1. Sensor nodes → CH (short-range scheduled uplink).
2. CH → Gateway (mesh/LoRa/wired).
3. Gateway → Cloud / Local Server (Ethernet/Wi-Fi/Cellular).
4. Server → Dashboard / Mobile alerts / Control commands.

## 7. Diagram of Designed WSN



## Conclusion

This greenhouse WSN design provides a scalable, energy-efficient, and reliable monitoring and control framework. The hybrid cluster-based mesh topology with TDMA/TSCH + LEACH (or RPL) balances energy use and robustness, while CHs and a central gateway provide aggregation, actuation, and cloud connectivity for advanced analytics and control.